CRI Exercise 09

1.2

It shows that the outliers either have been suppressed by the anonymization via ARX, or they have been binned such that there are no outliers anymore. So its not possible to have real outliers due to the binning. Furthermore, the distribution is significantly more compressed then the anonymized data in any visualization, this is also due to the binning.

2.2

The hypothesis tests with both the anonymized and anonymized data show that there is a positive association between average PM2.5 exposure in 2024 and Asthma reason encounters. The unanonymized data shows a Spearman correlation coefficient of $r \approx 0.457$ and the anonymized data a Spearman correlation coefficient of $r \approx 0.523$. Therefore the anonymized data shows a more strong correlation between asthma and the air quality patients live in. It also shows that the anonymzation by suppression and binning made the correlation go up between both variables. This is probably caused by removing outliers and extreme cases which may result in a stronger association. But this comes at the cost of reduced data granularity and may reflect an artefact of the anonymization rather than a real stronger relationship in the underlying population.